SELF E HACKATHON

2026

SMART MEMORY NECKBAND

CodeX



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PROBLEM

People with memory loss often forget familiar faces and important conversations, making everyday interactions difficult.

1. Pain Points

* Forget familiar people
* Cannot recall important conversations, appointments, or instructions
* Miss daily information such as schedules, medication, and reminders

2. Our Solution

A privacy-first, offline-capable smart neckband that:

* Recognises familiar people
* Remembers key conversations
* Provides timely reminders and important alerts

3. Existing Gaps

* Most solutions focus only on location tracking
* They do not help users remember people or conversations
* They require frequent user input and caregiver monitoring

4. Our USP

* Privacy-first design that works offline
* No internet required for core features
* Memory assistance in a single wearable device
* Simple and easy to use

5. Why It Matters

* Greater independence
* Peace of mind for families
* Safer daily life
* More confident social interactions



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SOLUTIONS

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1. SOLUTION WORK FLOW

Step 1: Capture Surroundings
* Continuously captures faces and speech using a camera and microphone.
* Uses voice activity detection to determine when to listen.

Step 2: On-Device Processing
* Recognises familiar faces.
* Converts speech to text.
* Summarises important conversations.
* Identifies known people.
* Processes data directly on the device, even without internet

Step 3: Secure Local Storage
* Stores recognised identities and conversation summaries locally.
* Uses encrypted storage.
* Applies time-limited data retention.
* Keeps data under the user's control.

Step 4: Real-Time Assistance
* Provides voice reminders about people.
* Recalls past conversations.
* Gives daily summaries directly on the device.
Example: “This is your granddaughter, Ananya. You last spoke about her college.”

Step 5: Caregiver Alerts and Access
* Shares key summaries and notifications with caregivers through an app when connected.
* Allows caregivers to view recognised people and conversation summaries.
* Provides important alerts, activity insights, and privacy settings.
* Syncs data optionally when an internet connection is available.

2. TECHNOLOGICAL APPROACH

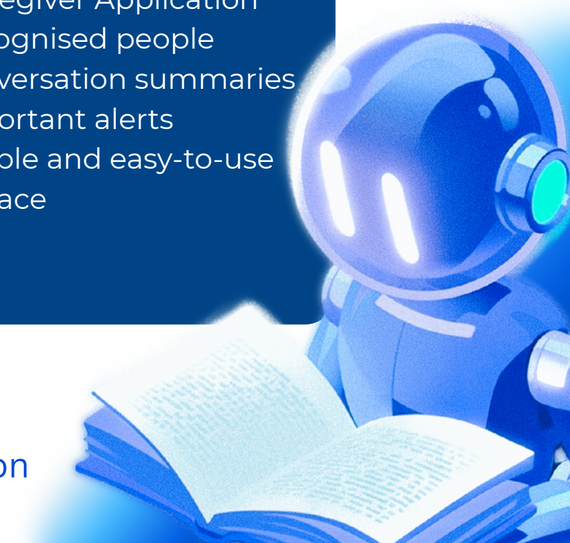
1. Hardware and Sensors
* Camera – Face recognition
* Microphone – Speech capture
* Microcontroller – On-device AI processing
* Battery – Portable power supply

2. On-Device AI Models
* Face recognition – OpenCV and FaceNet
* Speech-to-text – Whisper
* Conversation summarisation – Lightweight LLM
* Person matching – Known contacts

3. Data Management
* Encrypted local storage
* Time-limited data retention
* User-controlled privacy

4. Alerts and Communication
* Voice reminders through the neckband
* Caregiver app notifications when connected
* Optional daily summary synchronisation

5. Caregiver Application
* Recognised people
* Conversation summaries
* Important alerts
* Simple and easy-to-use interface



Key Highlight: The smart neckband provides personalised memory assistance while prioritising privacy, offline functionality, and ease of use.

Tools and Frameworks

- Python
- OpenCV – Face recognition
- Whisper – Speech-to-text
- Lightweight LLM – Conversation summarisation
- SQLite – Local storage
- MQTT/HTTPS – Optional cloud synchronisation

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Feasibility And Viability

Our smart neckband is technically achievable using readily available hardware components and open-source AI models.

Sensing
* Camera, microphone, and pulse sensor
* IMU (motion sensor)
* Optional GPS
* Compact, wearable, low-power components

On-Device AI
* Face recognition
* Speech-to-text
* Conversation summarisation
* Open-source models that work offline without paid APIs

Software
* Local encrypted storage
* Optional cloud synchronisation
* Caregiver application
* Privacy-first architecture
* Modular, scalable software that works without the internet

Our solution addresses the growing need for memory assistance and elderly care.

Target Users
* Elderly individuals with memory loss
* People in the early stages of dementia
* Caregivers and family members
* Assisted living homes and care centres

Phase 1: Prototype Development (0–3 months)
* Build core hardware
* Implement on-device AI
* Develop the basic neckband
* Build local storage and the app MVP

Phase 2: Testing & Validation (3–6 months)
* Conduct user testing with the target group
* Improve accuracy and usability
* Refine hardware design
* Evaluate privacy and safety

Phase 3: Pilot Deployment (6–12 months)
* Deploy in care homes and NGOs
* Gather real-world feedback
* Iterate and optimise
* Prepare for certification

Phase 4: Commercial Launch (1–2 years)
* Scale production
* Expand to healthcare providers
* Add advanced features
* Explore partnershipst



Conclusion:
The smart neckband combines available technology, affordable prototyping, privacy-focused AI, and meaningful social impact to support people with memory loss and their caregivers.



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IMPACTS AND BENEFITS

For Users

For Caregivers

For Society

For Healthcare Systems

01. For Users
Remember familiar people.
Recall important conversations.
Manage daily routines with voice reminders.

02. For Caregivers
Stay informed about daily activities and interactions.
View summarised conversations.
Customise person labels and reminders.



03. For Society
More independent and confident elderly population.
Stronger family and community bonds.
Creates opportunities for accessible, affordable assistive technology.

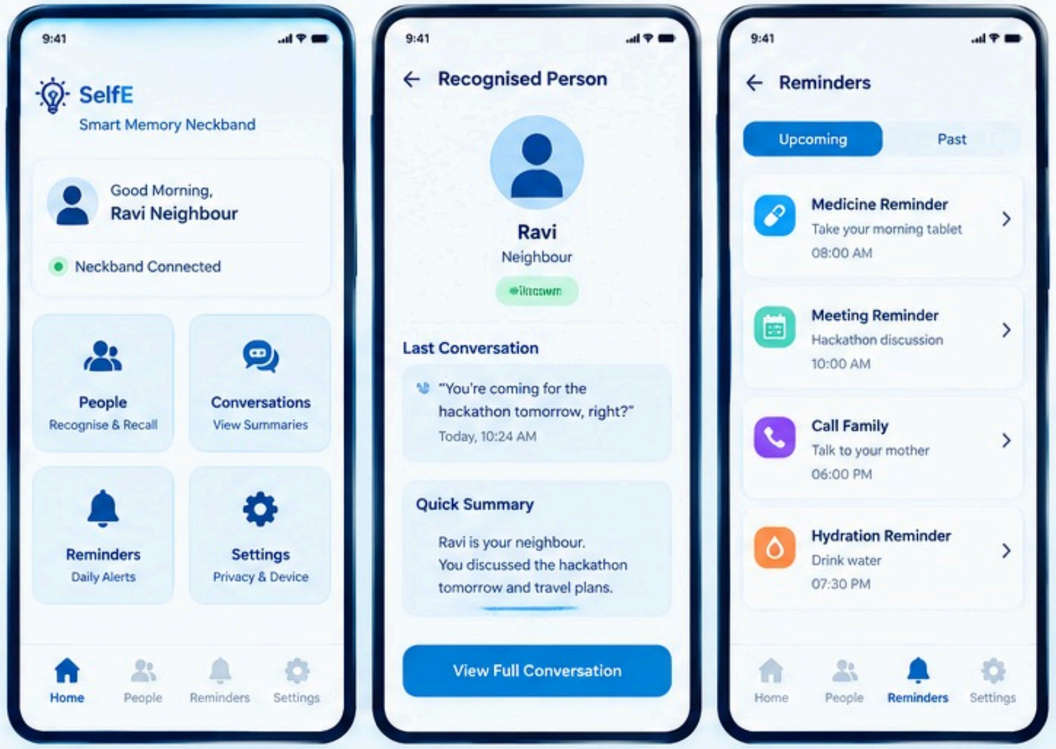


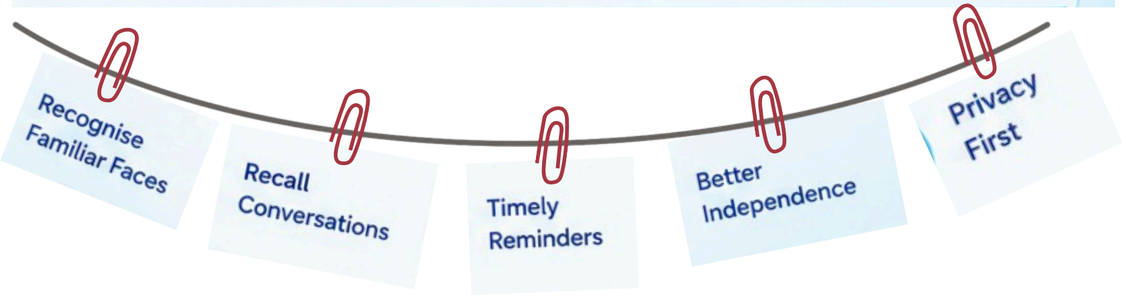
04. For Healthcare Systems
Early awareness of cognitive changes (via conversation patterns).
Supports timely intervention and follow-up.
Can reduce long-term care burden.

CIRCUIT DIAGRAM AND APP INTERFACE

APP INTERFACE (Mobile / Web)

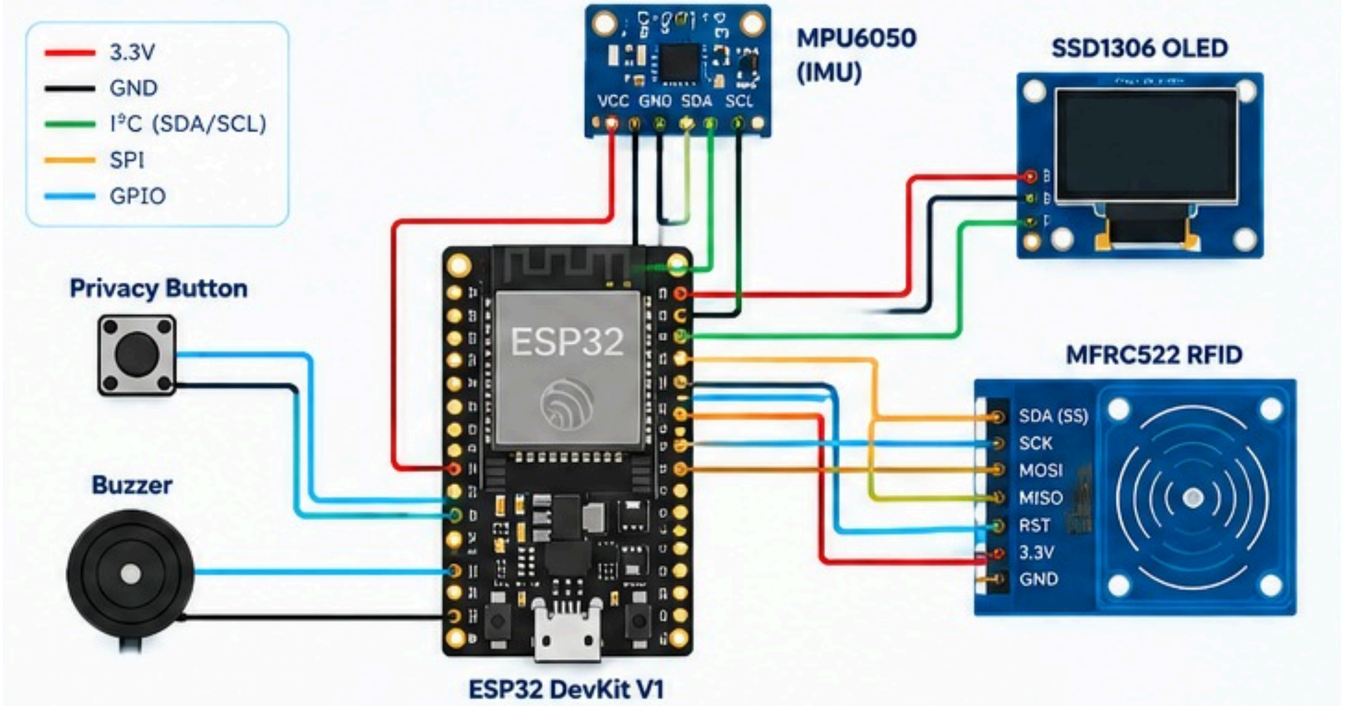
Simple. Intuitive. Accessible.





CIRCUIT DIAGRAM & WIRING

ESP32 + Sensors + Display + Alerts



Pin Mapping

MPU6050 (I²C)

VCC	→	3V3
GND	→	GND
SDA	→	GPIO 21
SCL	→	GPIO 22

OLED (I²C)

VCC	→	3V3
GND	→	GND
SDA	→	GPIO 21
SCL	→	GPIO 22

MFRC522 (SPI)

SDA (SS)	→	GPIO 5
SCK	→	GPIO 18
MOSI	→	GPIO 23
MISO	→	GPIO 19
RST	→	GPIO 27
3.3V	→	3V3
GND	→	GND

Others

Button	→	GPIO 26
Buzzer	→	GPIO 25 (to GND)

RESEARCH & REFERENCES

Sources and open-source work that shaped our design

1 CIRCUIT DIAGRAM

Wokwi Studio

ESP32 + sensor wiring simulated before building

wokwi.com

2 HARDWARE

Random Nerd Tutorials

ESP32, sensor and OLED setup guides

randomnerdtutorials.com

3 SQLITE / SQL

W3Schools

Queries for local people and summary storage

w3schools.com

4 REAL-LIFE DATA

WHO Dementia Facts

Prevalence and impact figures

who.int/news-room/fact-sheets/detail/dementia

Recognise
Remember



Protect
Empower

Thank *You*

